

1. Prototype chatbot:

Plan Your Route

Select your starting point, destination, and preferred pathfinding algorithm

Start Location

Hostel Block-1

Destination

Academic Block 2 (B)

Search Algorithm

A* ...

Find Optimal Path

A* Results

Route:

Hostel Block-1 → Food Court → Academic Block 2 (B)

Distance

150 meters

Est. Time

1.8 min

Explored

3 nodes

Exploration Order:

1. Hostel Block-1 2. Food Court 3. Academic Block 2 (B)

2. Working database and connection:

A working database connection : our chatbot is linked to a database that stores all campus details like: building names and distances between them . Instead of giving fixed answers, the bot fetches real information from the database whenever a student asks something. For example, if someone asks directions for other buildings, the chatbot checks the database and replies with route with directions. This makes the system dynamic, easy to update, and reliable.

3. Navigation integration:

Navigation integration in our project means combining the **campus graph** (buildings and paths) with the **search algorithms** (BFS, DFS, UCS, A*) and the **database**. The chatbot first understands the user's query, for example:

"Find path from Hostel Block-1 to Library."

Once the query is received, the navigation module uses the graph and chosen algorithm to calculate the **best route, total distance, and walking time**. At the same time. (The chatbot also connects to the database to fetch extra details about the destination, such as its type and opening hours.)